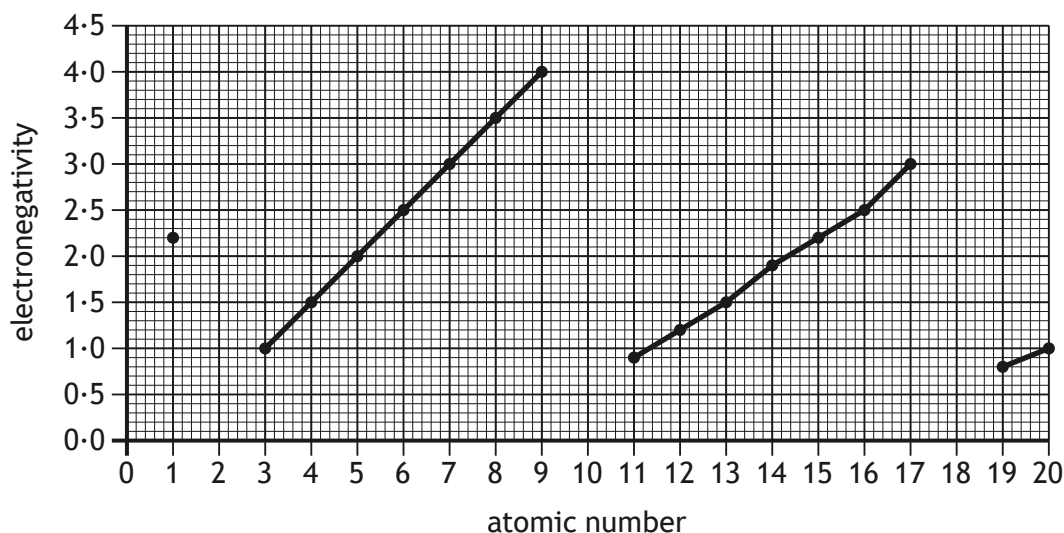


Total marks — 95
Attempt ALL questions

1. Electronegativity is a measure of the attraction an atom involved in a bond has for the electrons of the bond.

The graph shows the trend in electronegativity for the first 20 elements.



- (a) (i) State the trend in electronegativity as you go across period 2 from lithium to fluorine.

1

- (ii) No electronegativity values are shown for the elements with atomic numbers 2, 10 and 18.

Suggest why no values are provided for these elements.

1



* X 8 1 3 7 6 0 1 0 2 *

1. (a) (continued)

- (iii) On descending Group 7 from fluorine to iodine, the electronegativity of the elements decreases.

Explain why the electronegativity of the elements decreases as you go down the group.

1

[Turn over]

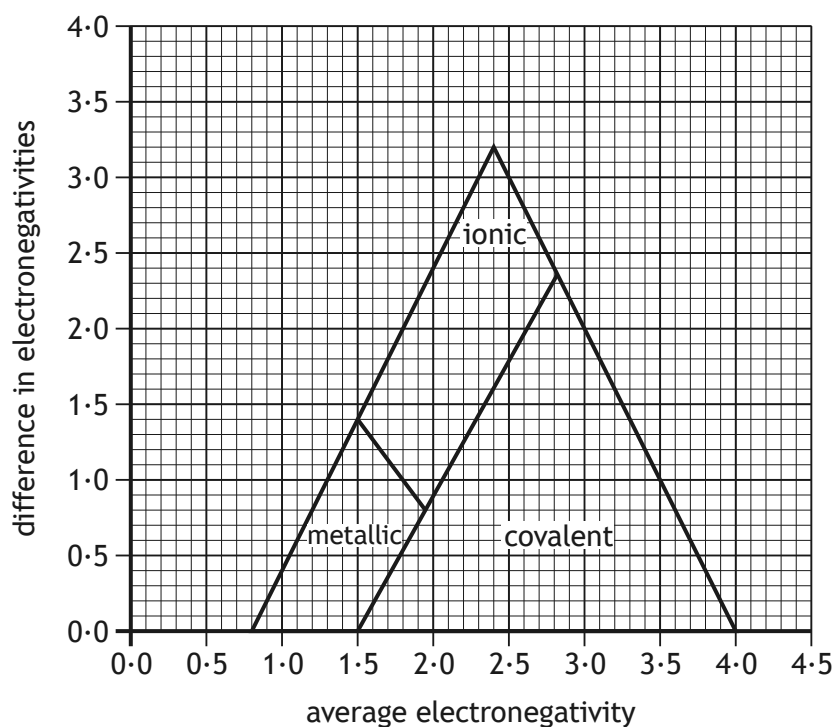


* X 8 1 3 7 6 0 1 0 3 *

1. (continued)

- (b) Electronegativity values can be used to predict the type of bonding present in substances.

The type of bonding between two elements can be predicted using the diagram below.



- (i) Using the information in the diagram, state the highest average electronegativity found in **ionic** compounds.

1



* X 8 1 3 7 6 0 1 0 4 *

1. (b) (continued)

MARKS

DO NOT
WRITE IN
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- (ii) The electronegativity values of magnesium and nitrogen are shown.

Electronegativity of magnesium = 1.2

Electronegativity of nitrogen = 3.0

Draw an X on the diagram on *page 04* to show the position of magnesium nitride.

Show your calculations clearly.

(An additional diagram, if required, can be found on *page 38*.)

2

- (iii) Compounds with a difference in electronegativities of 1.5 can have ionic or covalent properties.

- (A) The electronegativity difference between the elements in lithium sulfide is 1.5.

Write an **ionic** formula for lithium sulfide.

1

- (B) A compound contains two non-metal elements with an electronegativity difference of 1.5.

Suggest names for the two **non-metal** elements.

1

- (c) Fluorine has a greater attraction for bonding electrons than hydrogen.

State the term used to describe the type of **covalent** bond in hydrogen fluoride.

1



* X 8 1 3 7 6 0 1 0 5 *